

VLAN Trunking Protocol (VTP)

Project Overview

This project was initiated to explore the functionality and role of VLAN Trunking Protocol (VTP) in Cisco-based networks. The main goal is to understand how VTP simplifies VLAN management by synchronizing VLAN configurations across multiple switches within the same domain. Key objectives include:

- Understand how VTP uses trunk links and multicast to distribute VLAN info (IDs, names, revision numbers).
- Learn the roles of VTP modes: Server, Client, and Transparent.
- Examine the structure and types of VTP messages.
- Review configuration requirements: matching version, domain name, password, and trunking.
- Analyze the use of VTP pruning to reduce unnecessary VLAN traffic.
- Identify potential risks and affected locations when VTP is misconfigured.
- Apply best practices such as using VTP passwords, backups, and limiting server roles.

Description

VTP in Networking or VLAN Trunking Protocol is a Cisco proprietary protocol used by Cisco switches to synchronize VLAN information across a network. With VTP, network managers can handle VLANs on a single switch and reflect those changes on all other switches in the same VTP domain. It uses trunk links to send and receive VTP messages containing information such as VTP version, domain name, configuration revision number, and VLAN ID and name.

Recommendations

- Use VTP Transparent Mode
- Set a VTP Domain Name
- Use VTP Password
- Limit VTP Version
- Avoid Unintended VTP Server
- Backup VLAN Configurations
- Use VTP Pruning

References

<https://www.cisco.com/c/en/us/support/docs/lan-switching/vtp/10558-21.html>
<https://www.geeksforgeeks.org/computer-networks/vlan-trunking-protocol-vtp/>
<https://www.nwking.com/what-is-vtp>
<https://www.pynetlabs.com/what-is-vtp/>

Details

VLAN Trunking Protocol works by sending VTP messages over trunk links to other switches in the same VTP domain. VTP messages can be of different types, such as summary advertisements, subset advertisements, advertisement requests, and join messages. Each VTP message contains a configuration revision number. When a switch receives a VTP message with a higher revision number than its own, it updates its VLAN database accordingly.

There are some requirements for VTP to communicate VLAN information between switches. These are:

1. The VTP version must be same on the switches user wants to configure
2. VTP domain name must be same on the switches
3. One of the switches must be a server
4. Authentication should match if applied

VTP packets are sent in either Inter-Switch Link (ISL) frames or in IEEE 802.1Q (dot1q) frames. These packets are sent to the destination MAC address 01-00-0C-CC-CC-CC with a logical link control (LLC) code of Subnetwork Access Protocol (SNAP) (AAAA) and a type of 2003 (in the SNAP header).

There are two techniques of VTP in networking, namely-

- *Frame Filtering:* The frame filtering method analyses information regarding each frame (MAC address or a layer-3 protocol) in the VTP domain.
- *Frame Tagging:* The frame tagging method sets a distinctive identifier in the header as it gets forwarded to the VTP domain.

VTP modes - There are 3 modes:

- *Server* - The switches are set to this mode by default. This mode allows you to create, add and delete VLANs. The changes you want to make should be done in this mode. Any changes that are done on this mode (on a particular switch) will be advertised to all the switches that are in the same VTP domain. In this mode, the configuration are saved in NVRAM.
- *Client* - In this mode, the switches receive the updates and can also forward the updates to other switches (which are in the same VTP domain). The updates received here are not saved in NVRAM so all the configuration will be deleted if the switch is reset or reloaded i.e the switches will only learn and pass the VTP summary advertisements to the other switches.

- *Transparent* - This mode only forwards the VTP summary advertisements through trunk link. The transparent mode switches can make their own local database which keep secret from other switches. The whole purpose of transparent mode is to forward the VTP summary advertisements but not to take part in the VLAN assignments.

VLAN Trunking Protocol has three main components: VTP domain, VTP pruning, and VTP advertisements.

VTP Advertisements (Messages)

VTP shares VLAN info between switches to keep configurations synchronized across the same VTP domain. It sends updates via multicast over trunk links.

- Multicast Address: 01-00-0C-CC-CC-CC
- VTP Advertisement Includes:
 - VLAN ID
 - VTP Domain Name
 - VTP Password
 - Configuration Revision Number
 - VLAN Configuration

VTP Configuration Requirements:

- Trunking must be enabled
- Same VTP domain name and password on all switches
- Matching VTP version on all devices

VTP Pruning:

- Reduces unnecessary broadcast/multicast traffic
- Improves bandwidth by stopping VLANs not needed on certain links
- VLAN 1 cannot be pruned
- Disabled by default on switches

Affected Locations

- Switches
- Trunk Links
- Client Switches
- Access Layer Switches
- Multicast Domains
- Edge Locations with Non-Pruned VLANs